

Project 4 - Simulation Report

Vivado 2025.1.1 | xc7z020c1g400-1 | 50 MHz | audit: 9 October 2026 (IST)

Test setup and coverage

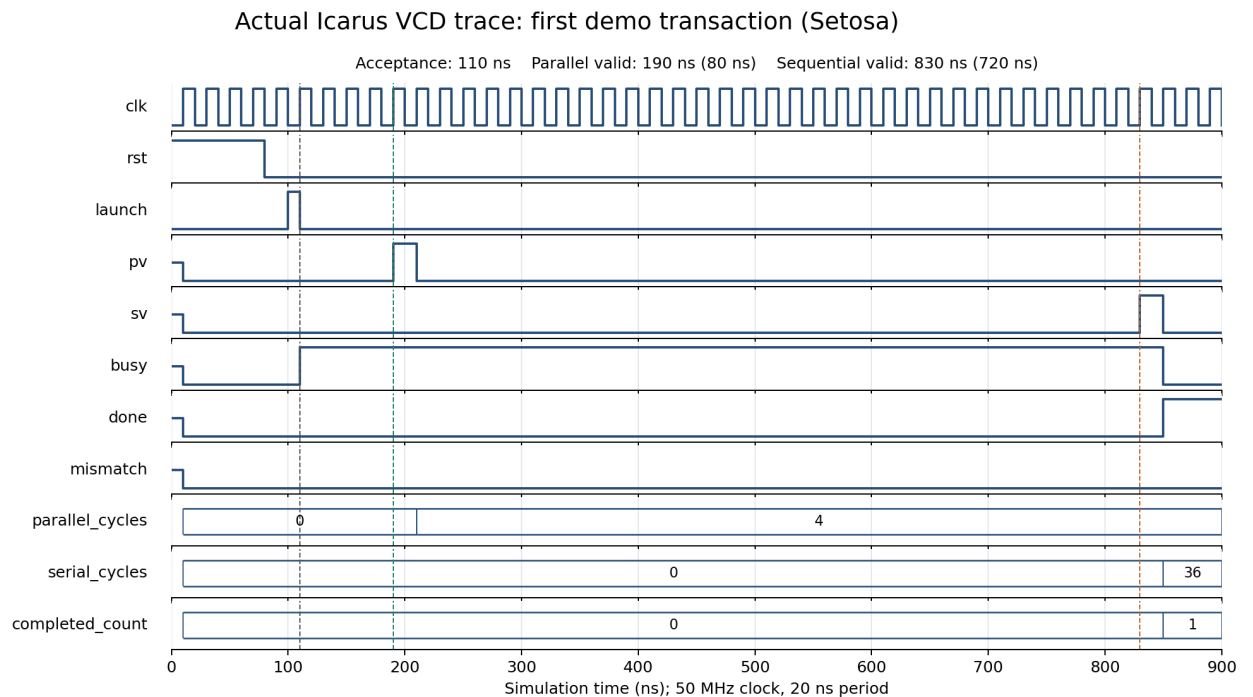
Recovered unchanged inference RTL, frozen model headers and 4,502 precomputed Python golden vectors. Simulator: Icarus Verilog 12.0. DUTs: nn_parallel, nn_serial and nn_demo_controller. Simulation clock period is 20 ns. No vendor VIO/clock IP or physical board is simulated by these benches.

Check	Observed result
Parallel reference suite	4,502 exact score/class matches; 4 elapsed periods
Sequential reference suite	4,502 exact score/class matches; 36 elapsed periods
Streaming / input latching	Consecutive parallel inputs and bubbles pass; serial captures input
Controller suite	150 Iris cases, hold/busy/reset handling, counters and mismatch pass
Arithmetic/reset boundaries	Reset abort/recovery, argmax ties and ReLU saturation pass
Three-case trace bench	All three exact reference scores; 80/720 ns valid timing

Actual regression output

PASS reset abort/recovery, argmax ties, ReLU saturation
PASS parallel: 4502 vectors, full-rate bursts + bubbles, latency=4 periods
PASS wrapper: 150 Iris samples, held toggle, busy rejection, reset abort, counters
PASS serial: 4502 vectors, input latching, latency=36 periods
ALL TESTS PASSED
Testbench/tb_nn.sv:169: \$finish called at 3421600000 (1ps)
The original Vivado/XSim log also contains ALL TESTS PASSED and is preserved in Results/vivado_batch.log. The full regression and waveform bench passed again in local Vivado XSim 2025.1.1 on 9 October 2026 (IST); the current_xsim files preserve these new outputs. Icarus outputs are separately named. Compiler executable files and temporary simulator files are excluded from the repository.

Measured waveform



Plot rendered from the genuine included VCD, not from expected values. Acceptance / parallel valid / sequential valid occur at 110 / 190 / 830 ns for class 0, then 1050 / 1130 / 1770 ns for class 1, and 1990 / 2070 / 2710 ns for class 2. Latencies are 80/720 ns in each case.

Supplementary capture output

VCD info: dumpfile Simulation/waveform.vcd opened for output.

Parallel core result at 191000: class=0 latency=4 cycles

Serial core result at 831000: class=0 latency=36 cycles

PASS demo d6d528e0: class=0 scores=19058,2857,-24757 cycles=4,36 count=1

Parallel core result at 1131000: class=1 latency=4 cycles

Serial core result at 1771000: class=1 latency=36 cycles

PASS demo 09110b2a: class=1 scores=-8102,8917,-3790 cycles=4,36 count=2

Parallel core result at 2071000: class=2 latency=4 cycles

Serial core result at 2711000: class=2 latency=36 cycles

PASS demo 37281210: class=2 scores=-13784,-6177,16696 cycles=4,36 count=3

WAVEFORM TEST PASS: 3 cases; exact scores; 4/36 cycle latency

Testbench/tb_nn_waveform.sv:63: \$finish called at 2900000 (1ps)

Reproduction: from the experiment folder run python python/run_open_source_tests.py with Icarus installed, then python Documentation/build_report.py. The script creates fresh transcripts, pass marker and VCD. Vivado builds use RUN_PROJECT4.cmd and the original tb_nn.sv.